



SESSION 12

Analysis of Variance (ANOVA)

One-Way ANOVA · F-Distribution · Sum of Squares Partitioning · Post-Hoc Analysis

What is ANOVA?

The F-Distribution & F-Statistic

Sum of Squares Decomposition

One-Way ANOVA Formulation

Degrees of Freedom in ANOVA

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Post-Hoc Analysis (Tukey's HSD)

Python Code Implementation

Session Notes — Complete Reference

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01 • What is ANOVA?

Analysis of Variance (ANOVA) is a statistical test used to compare the means of **three or more independent groups** simultaneously. It avoids the increased Type I error rate that occurs if we perform multiple pairwise t-tests.

02 • The F-Distribution & F-Statistic

ANOVA uses the **F-statistic**, which is the ratio of variability between groups to variability within groups:

F-STATISTIC FORMULA

$$F = \text{Mean Square Between (MSB)} / \text{Mean Square Within (MSW)}$$

If $F > 1$ and the p-value $\leq \alpha$, we reject the null hypothesis that all group means are equal.

03 · Sum of Squares Decomposition

ANOVA divides the total variance in the data into distinct components:

PARTITION OF SUM OF SQUARES

$SST = SSB + SSW$ SST = Total Sum of Squares (overall variance) SSB = Sum of Squares Between groups (treatment effect) SSW = Sum of Squares Within groups (error/residual variance)

04 • One-Way ANOVA Formulation

Null Hypothesis (\$H_0\$): $\mu_1 = \mu_2 = \dots = \mu_k$ (all group means are equal)

Alternative Hypothesis (\$H_a\$): At least one group mean is different from the others.

05 • Degrees of Freedom in ANOVA

- **Between Groups (df_{between}):** $k - 1$ (where k is the number of groups).
- **Within Groups (df_{within}):** $N - k$ (where N is the total sample size across all groups).

06 • ANOVA Assumptions

ANOVA assumes:

1. **Normality:** The residuals (errors) within each group must be normally distributed.
2. **Homogeneity of Variance:** All groups must have approximately equal variances (homoscedasticity).
3. **Independence:** Observations are independent.

07 • Post-Hoc Analysis (Tukey's HSD)

ANOVA is an **omnibus test** — it only tells you if *at least one* group is different, but not *which* ones. If ANOVA is significant, we perform **Tukey's Honestly Significant Difference (HSD)** test to check all pairwise comparisons.

08 · Python Code Implementation

```
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd

# Group data
g1 = [10, 12, 15, 11, 13]
g2 = [18, 20, 22, 19, 21]
g3 = [14, 15, 17, 16, 15]

# Perform One-way ANOVA
f_stat, p_val = stats.f_oneway(g1, g2, g3)
print(f"F-statistic: {f_stat:.4f}, p-value: {p_val:.4f}")
```

09 · Quick Reference Card



ANOVA Quick Reference

- **Use Case:** Comparing means of 3 or more groups.
- **F-Ratio:** Variance between groups / Variance within groups.
- **Post-Hoc:** Use Tukey's HSD only if the overall ANOVA is significant.